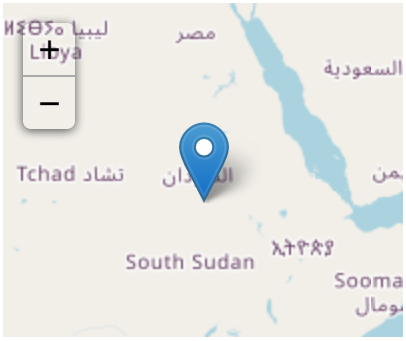
AquaPedia Case Study Database

Creative Options and Value Creation to Address Water Security in the Eastern Nile Basin

Case Description



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Geolocation:	12° 51' 46.11" N, 30° 13' 3.49" E
Total Population:	million
Total Area:	km^2
Climate Descriptors:	Arid/desert (Köppen B-type); Monsoon
Predominent Land Use Descriptors:	agricultural- cropland and pasture, conservation lands, urban- high density
Important Uses of Water:	Agriculture or Irrigation,Domestic/Urban Supply,Hydropower Generation,Other Ecological Services

Summary

Population increase and climate change are major concerns in the Eastern Nile Basin (ENB), North Africa. A proper management of water resources is crucial to prevent water crisis scenarios. Transboundary water (TBW) management require negotiations where parties recognize each other's interests and engage in joint fact-finding and the generation of mutually beneficial options. Options are to be packaged and "traded" among parties along the agreement delineation. Successful agreement, judged by its duration and compliance degree, have to be contingent and adaptive, accounting so for all forms of uncertainty. This paper addresses the issues of water quantity and water uses in ENB, looking at possibilities for creating and "packaging" water-related options, and the development of non-conventional water resources (NCWR). I found many water-related negotiations were carried out in the ENB during the 20th and first years of the 21st century, but none of them have efficiently settled the problems. Negotiation efforts have systematically violated international law principles; in addition, never were all interested/affected parties included in the negotiations rounds.

This stands clearly against the “active recognition” concept claimed by the Water Diplomacy Framework as an essential condition to enable successful TBW agreements. The Eastern Nile Basin has potential for the use of NCWR, which in turn might positively feed the options generation stage in future agreements. The Declaration of Principles signed in March 2015 is undoubtedly a starting point towards a comprehensive agreement in the whole Nile Basin. Appointing a third party neutral to reconcile the backstage of this focused agreement with the mission and achievements of the intergovernmental Nile Basin Initiative would be appropriate.



Cairo, Egypt

Natural, Historic, Economic, Regional, and Political Framework

Overview

This research is developed in the Eastern Nile Basin (ENB), Africa, where several aspects raise concerns in the international community and local stakeholders in terms of water resources availability and scenarios of potential water crisis. Population increase and climate change are often flagged as major concerns, but unfortunately not the only ones that need to be addressed. One way of understanding the water situation in the region is through the five indicators on water-quantity-related risks developed by the World Resources Institute (Gassert, Reig, Luo, & Maddocks, 2013). These five indicators range from 1 to 5 (low to high risk), and countries and regions rank differently depending on the indicator. For example, Egypt is at high risk before drought severity, whereas Ethiopia and Sudan are at high risk regarding seasonal variability and flood occurrence. ENB as a whole, however, is at low risk in terms of baseline water stress. Other relevant figures help depict the region in terms of socio-economic development and most of them hit the red zone of any colorful visualization. According to United Nations estimates, the total population in the Eastern Nile (EN) countries[1] increased by 50% between 2000 and 2015, overpassing 487 million. Yet, EN countries are not at the best in terms of welfare and socio-political stability. Most of them are low-income or lower middle-income countries according to the World Bank standards, show the lowest values for Human Development Index based on United Nations’ classification, and stability and social pressures pose these countries under the ‘alert’ flag according to the Fragile State Index by the Fund for Peace. Climate change projections are not clear in the Nile Basin. They are consistent in terms of rising temperatures, but not in terms of rainfall. North Africa might become hotter and drier, with an

increase in the evaporation component of the hydrologic budget (Conway, 2005; Islam & Susskind, 2015). Changes in hydrology equilibrium might not be good match for the increasing pressure posed in freshwater by current trends in economic development and land use patterns (Islam & Susskind, 2015), for instance the crop intensification practices (MWRI, 2014) and the land reclamation program in Egypt (FAO-AQUASTAT, 2015). “Water is clearly a major factor in socio-economic recovery and development in Africa. The continent appears to be blessed with substantial rainfall and water resources. Yet, it has severe and complex natural and man-made problems that constrain the exploitation and proper development of its water resources potential” (UN Water/Africa, N.D.)



Aswan, Egypt

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Ethiopian Highlands



Khartoum and neighbor cities, Sudan



Sudd, South Sudan

Both climate and context realities pictured in the overview section are unlikely to step back in the short-term. However, the water crisis, whether current or incoming, needs to be addressed now, under a multidimensional approach. The list of stakeholders involved in the EN water issues is extensive, as water cuts through most elements of human life. However, it is possible to identify the major actors for each of the two main issues addressed through this research, namely water quantity and water for what. Both issues involve macro- and micro-level interventions that can be addressed at different pace. However, a holistic approach beyond mere jurisdictional boundaries should be pursuit.

[1] Considering Sudan and South Sudan together

Stakeholders Matrix

		Issues	
Stakeholders		1) Water quantity	2) Water-Food-Energy nexus
		NSPD variables	
Number	Description	water quantity, ecosystems, governance	assets, governance, values and norms

1	Egypt-National Government	The country relies on the Main Nile River, coming from Sudan. It wants to keep water share at Aswan High Dam. Signed Declaration of Principles with main aim of not causing "significant damage" on riparians' water use	The country needs water for energy production and irrigation. Most of water is used in agriculture sector to grow food for increasing population.
2	Ethiopia-National Government	It claims having the right to develop. This country is the source of the Blue Nile, main tributary for the Main Nile	The country wants to harness its hydropower potential and so boost industrialization to escape poverty. It is building the Grand Ethiopian Renaissance Dam (GERD) with capacity for 6000 MW
3	Sudan-National Government	The country wants to make use of its water share. This would be possible if GERD is finished as Sudan would have sufficient and timely water stored upstream	Water is needed to foster development of irrigation schemes mostly for food production
4	South Sudan-National Government	The new country demands participation in the water share agreed by former Sudan and Egypt. However, current civil war offsets this demand. Current priority is safe water access for displaced population	It wants to develop agriculture for internal demand. It also has potential to become the "bread basket" of North Africa and Middle East.
5	China-National Government	***	China is interested in the region. It offers hydropower expertise and provide financial assistance. In returns, it has access to natural resources (oil and minerals).

6	Irrigation districts/schemes	Farmers in Egypt rely on water diverted at Aswan High Dam. Sudanese farmers rely on both, diversion and seasonal floods. Both rely on surface water, mostly from the Blue Nile River	Water is used for growing food for both economic activity and subsistence.
7	Big cities (Cairo, Khartoum)	Meet demand in growing urban areas for domestic and industrial use	Secure supply of water and energy for neighbors, institutions and businesses
8	Rural communities	Sufficient and steady water supply subsistence economies and decent livelihoods	Sufficient and steady energy supply for subsistence economies and decent livelihoods
9	Nile Basin Initiative	NBI aims in evening the access and use of water resources for all riparian countries	NBI promotes integrated management, sustainable development, and harmonious utilization of the water resources of the Basin, as well as their conservation and protection for the benefit of present and future generations. Its duties depend on financial donors, the World Bank being an important one.
10	World Bank	It won't provide funding for water-quantity-related projects in upper riparian countries unless lower ones are in fully agreement	***
11	Humanitarian organizations (UN, NGOs)	UN aims in communities to have sufficient water for decent subsistence.	Water access and food security in civil-war-affected South Sudan is a major concern.

12	Environmental organizations (RAMSAR Committee, etc.)	Concerns about water-related developments in the Sudd, South Sudan	The environment should be considered within the "nexus".
13	Energy companies	Hydropower companies are interest in preserve monthly and annual river flows	All energy-related companies have interests in the new dynamics introduced in the region because of GERD and other hydropower facilities in agenda in Ethiopia, as well as armed conflicts in oil areas in Sudan and South Sudan
14	Water Utilities	Meet demand in growing urban areas for domestic and industrial use. Make a profit	***
15	Import/export-linked countries		Production of export goods requires energy and water. Import opportunities are related to socio-economic development of communities.

Issues and Stakeholders

Water quantity. Nature or management issues?

NSPD: Water Quantity, Ecosystems, Governance
Stakeholder Types: Sovereign state/national/federal government, Local Government, Supranational union, Development/humanitarian interest, Environmental interest

It is claimed North Africa ran out of renewable freshwater decades ago, and the available water resources in the region are insufficient to meet food requirements (Qadir, Sharma, Bruggeman, Choukr-Allah, & Karajeh, 2007). Meanwhile, the total annual rainfall in the entire Nile Basin averages 2,000 billion cubic meters (BCM) (Awulachew, 2012; Islam & Susskind, 2015), and less than 5%

makes it to Lake Nasser. There is a supply-demand gap in the region, likely as result of naturally uneven distribution of the water (hydrologic problem) combined with a mismanagement of the water resources (human problem).

The Nature/Social/Politics Domain (NSPD) variables associated to this issue are water quantity, ecosystems, governance. In turn, the corresponding stakeholders are local, state/province, and national governments within EN basin (Egypt, Eritrea, Ethiopia, Sudan, South Sudan); multi-national institutions (UN, World Bank, Nile Basin Initiative, RAMSAR Committee, etc.) development/humanitarian NGOs (Water.org, Oxfam, etc.), irrigation districts/schemes, water utilities, energy companies.

Water for what? The water-food-energy nexus.

NSPD: Water Quantity, Governance, Assets, Values and Norms
Stakeholder Types: Sovereign state/national/federal government, Local Government, Supranational union, Development/humanitarian interest, Environmental interest, Industry/Corporate Interest

The gap in water supply and demand is far from encapsulated. Quite the opposite, it is complexly intertwined with other social, political, and economical problems. More population implies more drinking water and more food as well; food whose production demands more water. Energy access is also a constraint in the region, particularly in the Sub-Saharan countries where 70% of people lack of access to reliable energy sources. Food and energy security turn out to be intimately related to water (water-food-energy nexus) when addressing the question of water either for irrigation and food production to fight poverty, or power generation to boost industrialization. The NSPD variables associated to this issue are assets, governance, values and norms. In turn, the stakeholders linked to the issue are national governments within EN basin (Egypt, Eritrea, Ethiopia, Sudan, South Sudan); import/export-linked foreign countries (EU, USA, China); multi-national institutions (UN, World Bank, Nile Basin Initiative, RAMSAR Committee, World Trade Organization) development/humanitarian NGOs (Water.org, Oxfam), corporate interests (irrigation districts/schemes, energy companies, oil-and-gas companies)

It is possible to shift the conversation focus, moving from the current one of “how to split what we have” to a new one about “how to increase the pie, rearrange and redistribute”. To do that, we need to look at the region as a whole, and explore the idea of “creating more water”, as defined in the Water Diplomacy Framework.

Analysis, Synthesis, and Insight

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Key Questions

Transboundary Water Issues: *What mechanisms beyond simple allocation can be incorporated into transboundary water agreements to add value and facilitate resolution?*

Creating and packaging options

The first research question is about how options or strategies, that address water issues creatively and/or build on possible technology innovations, can be packaged and employed to create non-zero sum choices within negotiations. Options need to be creatively generated and then packaged in such a way that the value of the whole is larger than the sum of the individual parts. This is the key of the value creation principle in negotiation. In the arena of the water resources management, competing interests can be met simultaneously if stakeholders find ingenious ways of using the same water in a variety of ways. This means "tearing apart" the traditional view that water is a fixed resource and introducing the concept of "water as a pie" that can be enlarged. Islam and Susskind (2012) argue that successful value creation requires time investment to "make the pie as large as possible" before distributing gains and losses. This is very much linked with the need of understanding each other's interests, for which information sharing is crucial. Even though the theory is quite clear and tempting, applications of this approach help the reader better understand its virtues in context. L. E. Susskind and Rumore (2015) tested the application of "devising seminars", originally developed by Fisher and others in the 70s, on Artic fisheries. The rationale behind this tool is bringing together stakeholders for an off-the-record, facilitated event were stakeholders brainstorm around the collective problems. The authors believe this tool helps overcome the barrier of lack of good and widely-supported ideas typically present in the public policy arena. The CALFED Delta-Bay case in California (USA) is one of the best-documented instances of water-negotiation. Multiple stakeholders participated, shifting their mission from a battle over who would make sacrifices to a search for new ways of managing the resource. Open information was a key element in the success of this case, as the availability of real-time information on water allowed stakeholders to better foresee and plan accordingly (Islam & Susskind, 2012). This model emerged after several years of frustrated negotiations under the traditional governance system, giving place to a new one where collaboration and adaptive management are central (Innes, Connick, Kaplan, & Booher, 2006) Gryzbowski, McCaffrey, and

Paisley (2009) analyzed several water treaties, particularly focusing on scenarios of and approaches for negotiations. Scenarios might be either “narrow” or “open”, depending on whether parties get stuck on mere definitions or they recognize their pros and cons and use the time to mutual gains developments. The approach to negotiate might be either positional or interest-geared. Provided most negotiations involve parties with more than one problem or concern, this is an opportunity for value creation. The treaty of peace between Israel and Jordan in 1994 is an example of value creation and trust enhancement. Parties included within the agreement elements such as desalination, water banking and transfer in water rights, showing that the combination of technological innovation and a collaborative administration can facilitate problem solving and enhance the chances for win-win sustainable solutions (L. Susskind & Islam, 2012). Conversely, the Danube River negotiation case in 1994 was wrongly addressed, as the two concerns for the riparian countries, economic development and environment, were addressed separately, missing the opportunity of value creation. “The results of the two negotiations cancelled each other out”, yielding winners and losers (Islam & Susskind, 2012) Value creation in TBW disputes happens when parties engage in joint-fact finding, formulate contingent agreements and emphasize adaptive management (Islam & Susskind, 2012). In addition, sustainable solutions are a consequence of well-designed problem-solving or negotiation process (Innes and Booher, 2010 in Islam and Susskind, 2012). Since 1999, the Nile Basin Initiative (NBI) has been working on these lines. Primarily funded by the World Bank, NBI has generated a significant critical mass of projects through its two Subsidiary Actions Programs. One example of this is the Eastern Nile Multi-Sectoral Investment Opportunity Analysis (EN-MSIOA), one of a set of specific studies being carried out to facilitate cooperative water resource management and development in the Nile Basin (ENTRO, 2014). This study intends to support strategic planning decisions at the scale of the ENB, through different scenario simulations. Another recent study examined benefit sharing opportunities among the riparian countries in terms of water resources management. The authors explored the potential shared benefits steaming from the development and operation of the Grand Ethiopian Renaissance Dam (GERD), under construction in Ethiopia, and its impact to downstream countries and the High Aswan Dam. They found there is possibility for Ethiopia to be better off without any country being worse off if these two dams are managed in coordination (Habteyes, El-bardisy, Amer, Schneider, & Ward, 2015) I can affirm there are institutions in place, a portfolio of projects, and the urgent need for sustainable long-standing solutions. What is then keeping the Nile River Basin from taking off? It is time to put options together, package and deliver. For this to happen, however, enabling conditions are needed as discussed later on.

Technological Innovation:

Non-conventional water resources

The second research question inquiries about the potential water increments/savings to be reached through the implementation of non-conventional water resources (NCWR). They can be used as a complement to conventional water resources to relieve water scarcity in regions where renewable

water resources are insufficient. Qadir et al. (2007) present an interesting review on NCWR with focus on food-security in Middle East and North Africa. Desalination of seawater and highly brackish groundwater, rainwater harvesting and the use of marginal-quality water resources for irrigation are mentioned as alternatives for water augmentation at different scales. ENB countries show little and scattered efforts on the road of NCWR. Desalination in Egypt has been given low priority because the cost of treatment is high compared with other sources (MWRI, 2014). Several desalination plants operate on the coasts of the Red Sea and the Mediterranean Sea to provide water for seaside resorts and hotels (FAO-AQUASTAT, 2015), where the value of water is high enough to cover the treatment costs (MWRI, 2014). Sudan started to walk the same road, provided seawater desalination has been recently introduced in Port Sudan town (FAO-AQUASTAT, 2015) In terms of rainwater harvesting and the reuse of drainage and wastewater, the difference among the countries is notorious. In Egypt, rainfall occurs only in winter and it cannot be considered a reliable source provided its high spatial and temporal variability (MWRI, 2014). Despite of this, rainwater harvesting is practiced in the regions of Matruh and North Sinai (FAO-AQUASTAT, 2015). On the other hand, traditional water harvesting practices are found in all the states of Sudan, where small reservoirs catch rainfall and runoff for domestic use in villages and pastoralists in remote areas (FAO-AQUASTAT, 2015). Concerning water reuse, about 25 to 30% of agricultural drainage of irrigation water in areas on both sides of the Nile Valley returns to the River Nile or main irrigation canals in Upper Egypt and in the southern Delta. Also, Egypt takes the lead in the region by treating part of its municipal wastewater and reusing it for irrigation either directly or mixed with drainage water (FAO-AQUASTAT, 2015; MWRI, 2014). The region shows potential for increasing, improving and introducing NCWR. As an instance on this regard, I look at good examples within and outside ENB. A small country, Singapore hosts 3.4 million people inside 720 square kilometers of territory (similar to New York City). Despite its lack of natural water resources and pollution in its rivers, this “big city” overcome water shortages by building a robust, diversified and sustainable water supply system out of four different sources known as the Four National Taps, referencing local rainwater harvesting, wastewater reuse, desalination technology and imported water (PUB, 2015) Singapore had mostly water quality problems due to discharges of raw wastewater from households and formal and informal economic activities (pig farms, shipbuilding industry, etc.). This forced the country to ration the water supply and thereby depend deeply on water transfer from the neighbor country Malaysia. In the period 1977-1987, eleven government agencies worked together to reverse an unsustainable reality. Today, the country collects water from 2/3 of its territory through an 8,000-Km drain network and storages it in 17 reservoirs. It also recycles wastewater using micro-filtration, reverse osmosis and ultraviolet disinfection, which is used to cover 30% of freshwater demand, mainly in the industry sector and for topping up reservoirs in dry seasons. In addition, Singapore uses reverse osmosis to turn seawater into freshwater, meeting 25% of the demand. Finally, the country’s fourth tap comes through an agreement with Malaysia, valid until 2061, which enables the island to withdraw up to 250 MGD from the Johor River (Tang, 2015) I believe there might be something to learn from Singapore that could be applied in the densely populated cities of Cairo and Khartoum, which host more than 18 and 5 million people respectively?

Tagged with: [[Topic Tag::Nile Basin|Topic Tag::Nile Basin, value, negotiation, enabling conditions, water resources'|water resources']]

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